

Def Fourier Series

$A_0, \{A_n\}_{n \in \mathbb{N}}, \{B_n\}_{n \in \mathbb{N}}$ be coeffs

$$A_0 + \sum_{n=1}^{\infty} \left(A_n \frac{\cos n\pi x}{L} + B_n \frac{\sin n\pi x}{L} \right) \text{ is a FS}$$

Thm For $\forall f: (-L, L) \rightarrow \mathbb{R}$ satisfying Direhlet's conditions

$\exists A_0, \{A_n\}_{n \in \mathbb{N}}, \{B_n\}_{n \in \mathbb{N}}$ s.t.

(1) finite discont. pts
(2) finite extreme pts

FS with coeff above

$$f(x) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\pi x + B_n \sin n\pi x), \quad x \in (-L, L)$$

Def Kronecker delta $\delta: \mathbb{N}^2 \rightarrow \{0, 1\}$

$$\delta_{mn} = \begin{cases} 1, & m=n \\ 0, & m \neq n \end{cases}$$

(to define orthogonality)

Thm Orthogonality of tripg functions

$$\forall m, n \geq 0, \quad \int_{-L}^L \cos n\pi x \cos m\pi x = \begin{cases} 2L, & m=n=0 \\ L\delta_{mn}, & \text{if } m, n \neq 0 \end{cases}$$

$$\int_{-L}^L \sin n\pi x \sin m\pi x = \begin{cases} 0, & m=n=0 \\ L\delta_{mn}, & \text{if } m, n \neq 0 \end{cases}$$

$$\int_{-L}^L \sin n\pi x \cos m\pi x = 0 \quad \forall m, n$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\cos(2x) = 1 - 2\sin^2 x = 2\cos^2 x - 1$$

$$\sin(2x) = 2\sin x \cos x$$

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

$$\tan\left(\frac{x}{2}\right) = \frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x}$$

$$\sin^2 x + \cos^2 x = 1$$

$$A \sin x + B \cos x = \sqrt{A^2 + B^2} \sin\left(x + \tan^{-1}\left(\frac{B}{A}\right)\right)$$

因而

$$\int_{-L}^L \cos n\pi x \cos m\pi x = \int_{-L}^L \frac{1}{2} \left(\cos\left(\frac{n-m}{2}\pi x\right) + \cos\left(\frac{n+m}{2}\pi x\right) \right)$$

$$\Rightarrow \text{Take } n \neq m, \int_{-L}^L \cos n\pi x \cos m\pi x = 0 \quad \text{if } n \neq m$$

$$\text{Take } n=m \neq 0, \int_{-L}^L \cos n\pi x \cos n\pi x = \frac{1}{2} \int_{-L}^L (\cos 0 + \cos 2n\pi x) = L$$

□

$$\forall m, n \Rightarrow \sin n \perp \cos m$$

$$f: [-L, L] \rightarrow \mathbb{R} \text{ s.t. } \int_{-L}^L f^2 < \infty$$

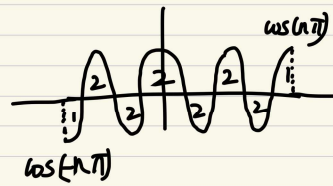
on $L^2([-L, L])$

$$\forall m \neq n \Rightarrow \sin n \perp \sin m, \cos n \perp \cos m$$

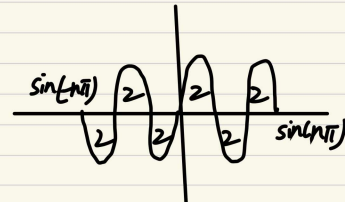
Pf

$$\forall n \neq 0, \int_{-L}^L \cos n\pi x = 0$$

$$(n=0 \Rightarrow \int_{-L}^L \cos n\pi x = \int_{-L}^L 1 = 2L)$$



$$\forall n, \int_{-L}^L \sin n\pi x = 0$$



Review: trigono eq.

积化和差公式: $\cos A \cos B = \frac{1}{2} (\cos(A-B) + \cos(A+B))$

$$\sin A \sin B = \frac{1}{2} (\cos(A-B) - \cos(A+B))$$

$$\sin A \cos B = \frac{1}{2} (\sin(A-B) + \sin(A+B))$$

(反过来, 和化积公式: $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

Thm Formula for Fourier coefficients

$$A_0 = \frac{1}{2L} \int_{-L}^L f$$

$$A_n = \frac{1}{L} \int_{-L}^L f \cos n\pi x$$

$$B_n = \frac{1}{L} \int_{-L}^L f \sin n\pi x$$

Pf: Using orthogonality

$$\int_{-L}^L f = \int_{-L}^L A_0 + \sum_{n=1}^{\infty} (A_n \cos n\pi x + B_n \sin n\pi x)$$

(我们 assume f Fourier series $\neq (-L, L)$ $\perp$ uni. conv.)

$$= \int_{-L}^L A_0 + \sum_{n=1}^{\infty} \left(\underbrace{A_n \int_{-L}^L \cos n\pi x}_{0} + \underbrace{B_n \int_{-L}^L \sin n\pi x}_{0} \right)$$

$$= \int_{-L}^L A_0 = 2LA_0$$

$$\Rightarrow A_0 = \frac{1}{2L} \int_{-L}^L f$$

Multiplying $\sin n\pi x$ on both sides to determine B_n ($n \neq 0$)

$$\int_{-L}^L f(x) \sin n\pi x = \underbrace{A_0 \int_{-L}^L \sin n\pi x}_{0} + \sum_{m=1}^{\infty} \int_{-L}^L (A_m \cos m\pi x + B_m \sin m\pi x) \sin n\pi x$$

$$= \sum_{m=1}^{\infty} B_m L \delta_{mn} = B_n L \quad \text{只当 } n=m$$

$$\Rightarrow B_m = \frac{1}{L} \int_{-L}^L f \sin_m$$

同理 by multiplying $\cos_m$ on both sides 可得

$$A_m = \frac{1}{L} \int_{-L}^L f \cos_m$$

Prop odd function 的 Fourier series 只有 sin 项
even function 的 Fourier series 只有 cos 项

$$f(x) = A_0 + \sum_{n=1}^{\infty} (A_n \cos_n + B_n \sin_n)$$

$$\text{for odd } f: A_n = \frac{1}{L} \int_{-L}^L f \cos_n \Rightarrow 0$$

$$\text{even } f: B_n = \frac{1}{L} \int_{-L}^L f \sin_n \Rightarrow 0$$

Complex form of Fourier series

recall: by Euler's formula, $e^{ix} = \cos x + i \sin x$

$$\Rightarrow \cos x = \frac{e^{ix} + e^{-ix}}{2}, x \in \mathbb{C}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}, x \in \mathbb{C}$$

($x \in \mathbb{R}$ 时 1 会抵消, 也是 closed 的)

$$\text{因而 } \cos_n = \cos \frac{n\pi x}{L} = \frac{e^{\frac{in\pi x}{L}} + e^{-\frac{in\pi x}{L}}}{2}$$

Thm Fourier series 的 complex form

$$f(x) = \sum_{n \in \mathbb{Z}} \alpha_n e^{\frac{in\pi x}{L}}$$

$$\text{where } \alpha_n = \frac{1}{2L} \int_{-L}^L f e^{\frac{-in\pi x}{L}}$$

$$\begin{aligned} \text{pf } f(x) &= A_0 + \sum_{n=1}^{\infty} (A_n \cos_n + B_n \sin_n) \\ &= A_0 + \sum_{n=1}^{\infty} \left(A_n \frac{e^{\frac{in\pi x}{L}} + e^{-\frac{in\pi x}{L}}}{2} + B_n \frac{e^{\frac{in\pi x}{L}} - e^{-\frac{in\pi x}{L}}}{2i} \right) \\ &= A_0 + \sum_{n=1}^{\infty} \left(\frac{A_n - iB_n}{2} e^{\frac{in\pi x}{L}} + \frac{A_n + iB_n}{2} e^{-\frac{in\pi x}{L}} \right) \end{aligned}$$

Define $\alpha_0 = A_0 \rightarrow \alpha_n \rightarrow \alpha_{-n} (n > 0)$

$$\Rightarrow f(x) = \sum_{n \in \mathbb{Z}} \alpha_n e^{\frac{in\pi x}{L}}$$

Rmk $\forall n \in \mathbb{N}, \alpha_n = \overline{\alpha_{-n}}$

推 α_n :

$$\alpha_0 = A_0 = \frac{1}{2L} \int_{-L}^L f = \frac{1}{2L} \int_{-L}^L f e^0 = \frac{1}{2L} \int_{-L}^L f e^{\frac{-i0\pi x}{L}}$$

for $n \in \mathbb{Z}_{>0}$,

$$\begin{aligned} \Rightarrow \alpha_n &= \frac{A_n - iB_n}{2} = \frac{1}{2L} \int_{-L}^L f \cos_n - \frac{i}{2L} \int_{-L}^L f \sin_n \\ &= \frac{1}{2L} \int_{-L}^L f (\cos_n - i \sin_n) \\ &= \frac{1}{2L} \int_{-L}^L f \cos \frac{n\pi(-x)}{L} = \cos \frac{n\pi(-x)}{L} \\ &= \frac{1}{2L} \int_{-L}^L f e^{\frac{-in\pi x}{L}} \end{aligned}$$

for $n \in \mathbb{Z}_{<0}, n = -m$ for some $m > 0$

$$\begin{aligned} \Rightarrow \alpha_n &= \alpha_{-m} = \frac{1}{2L} \int_{-L}^L f e^{\frac{im\pi x}{L}} \\ &= \frac{1}{2L} \int_{-L}^L f e^{-\frac{-im\pi x}{L}} \end{aligned}$$

$$\text{因而 } \forall n \in \mathbb{Z}, \alpha_n = \frac{1}{2L} \int_{-L}^L f e^{\frac{-in\pi x}{L}}$$

Thm orthogonality of complex Fourier series

$$\forall m, n, \int_{-L}^L e^{\frac{in\pi x}{L}} \cdot e^{\frac{-im\pi x}{L}} = 2L \delta_{mn}$$

Rmk Conjugate

$$\overline{a+bi} = a-bi$$

$$\text{那么指数形式: } e^{-i\theta} = \overline{\cos \theta + i \sin \theta} = \cos \theta - i \sin \theta = e^{-i\theta}$$

于是在 $L^2([-L, L])$ 上, 可以定义 inner product

$$\langle f, g \rangle = \int_{-L}^L f \bar{g}$$

因而 complex Fourier series 的 orthogonality.

$$L^2([-L, L]) \text{ 上, } e_n \perp e_m \text{ if } n \neq m$$

Fourier cosine & sine series

我们可以把一个 function extend 成为一个 even/odd function
使得新 function 的 Fourier series 只含有 sin/cos.

Def even extension

$$f_E(x) = \begin{cases} f(x), & 0 < x < L \\ 0, & x=0 \\ f(-x), & -L < x < 0 \end{cases}$$

称之为 Fourier cosine series

$$\Rightarrow f_E(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{L}, \quad -L < x < L$$

$$A_0 = \frac{1}{2L} \int_{-L}^L f_E = \frac{1}{L} \int_0^L f$$

$$A_n = \frac{1}{L} \int_{-L}^L f_E \cos_n = \frac{2}{L} \int_0^L f \cos_n$$

Def odd extension

$$f_o(x) = \begin{cases} f(x), & 0 < x < L \\ 0, & x=0 \\ -f(-x), & -L < x < 0 \end{cases}$$

称之为 Fourier sine series

$$\Rightarrow f_o(x) = \sum_{n=1}^{\infty} B_n \sin_n, \quad B_n = \frac{2}{L} \int_0^L f \sin_n$$

Thm Orthogonality relations of sine/cosine Fourier series

$$\int_0^L \sin_n \sin_m = \begin{cases} 0, & n=m=0 \\ \frac{L}{2} \delta_{nm}, & \text{otherwise} \end{cases}$$

$$\int_0^L \cos_n \cos_m = \begin{cases} L, & n=m=0 \\ \frac{L}{2} \delta_{nm}, & \text{otherwise} \end{cases}$$